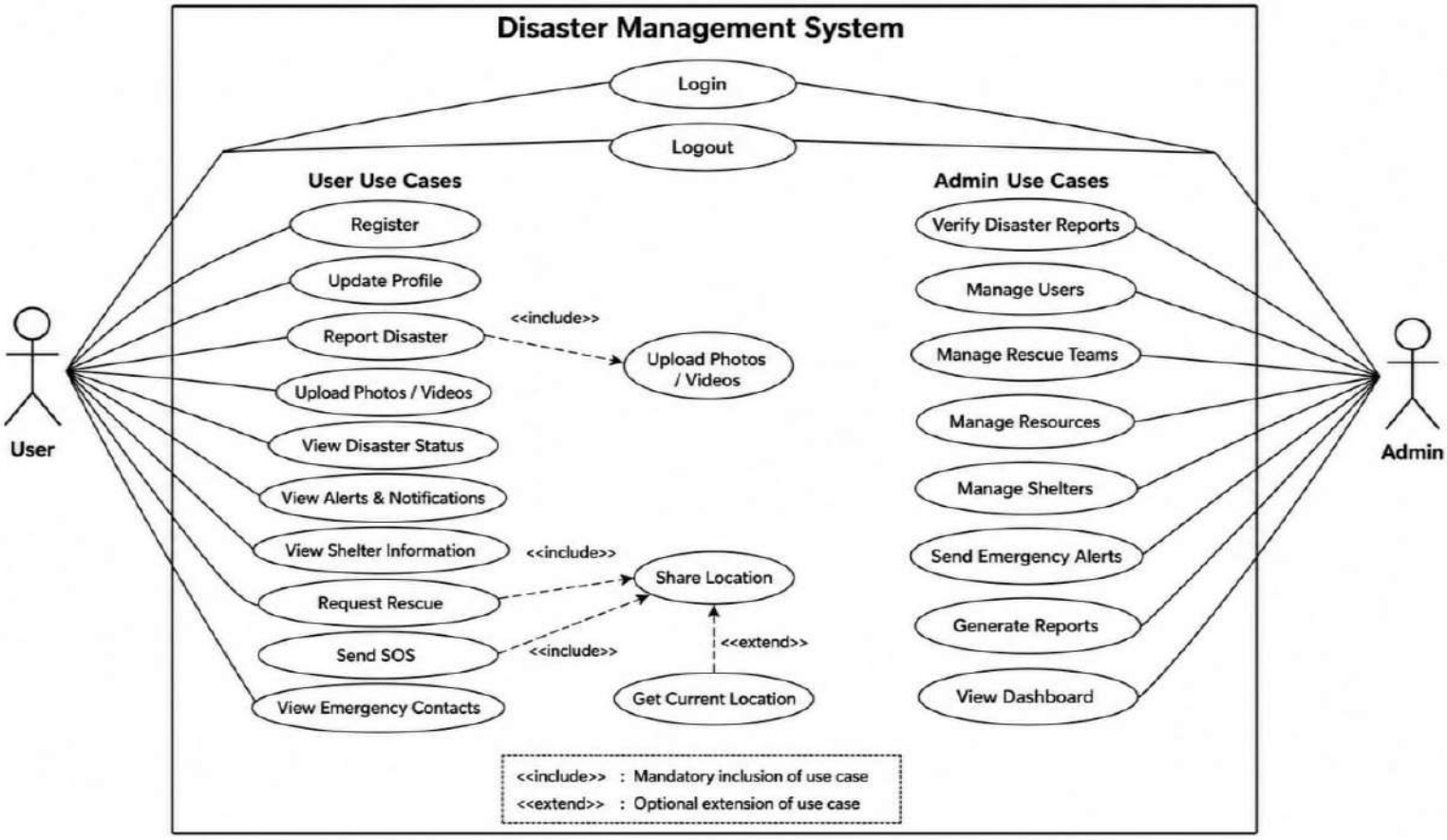
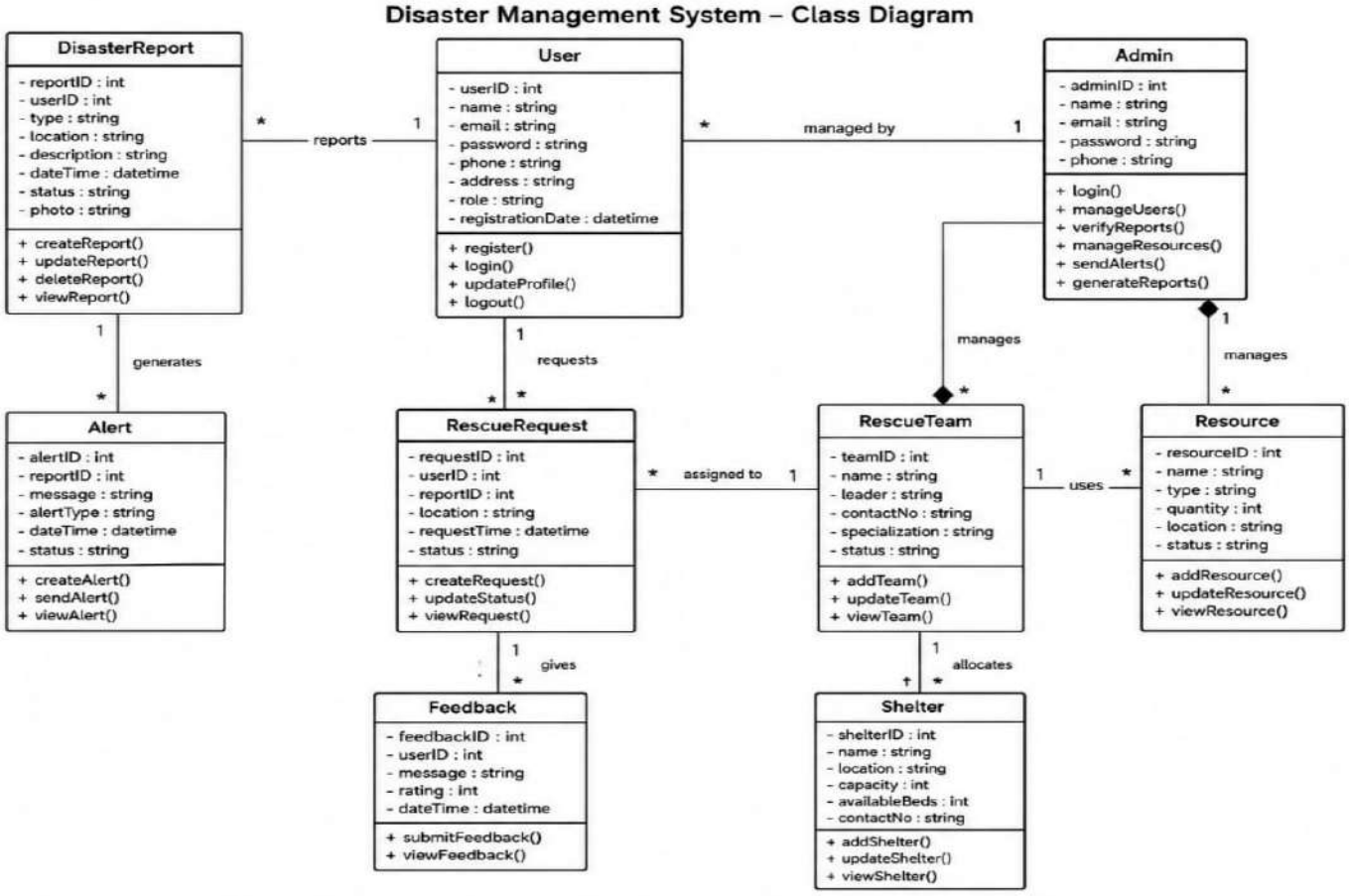
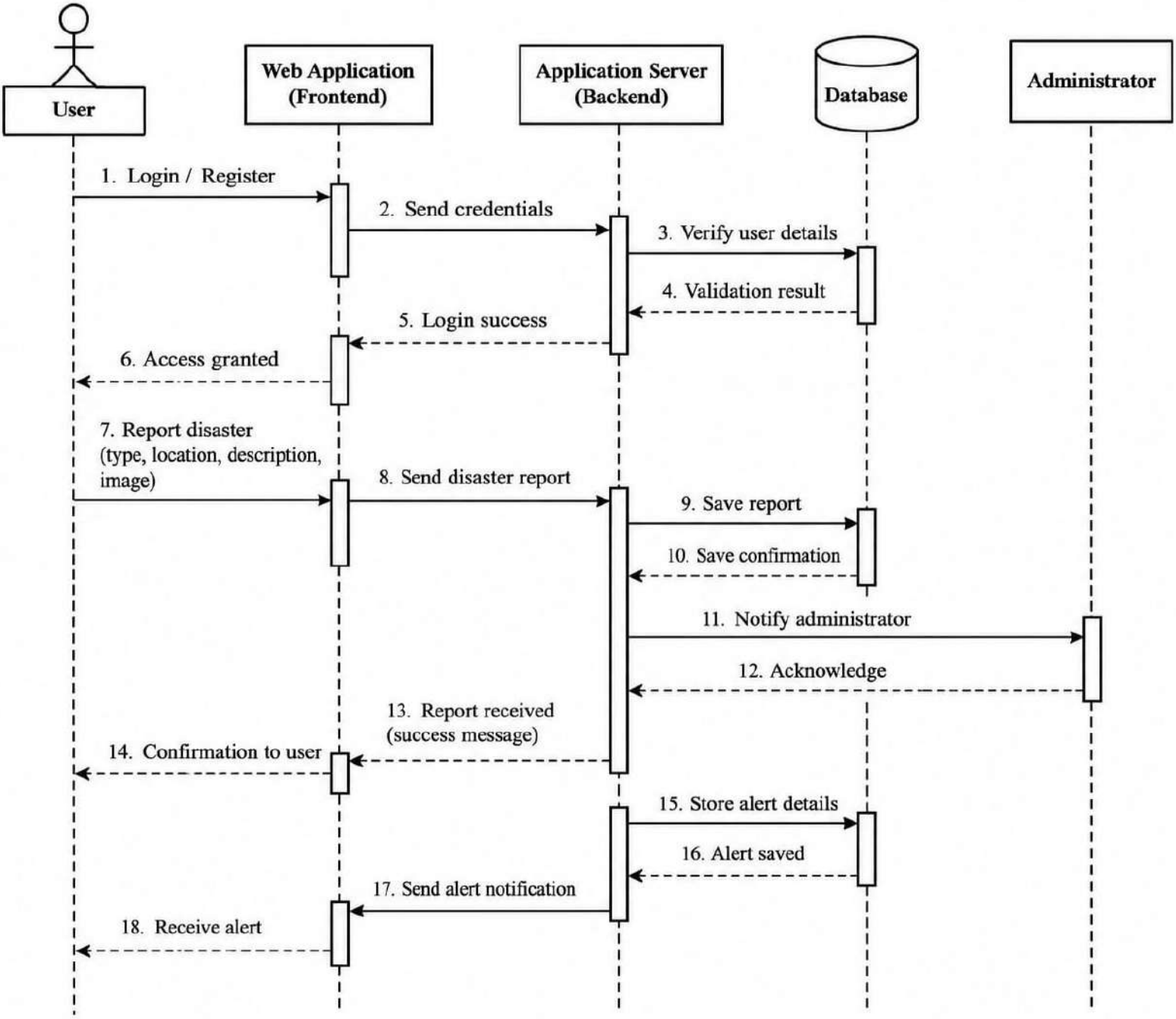
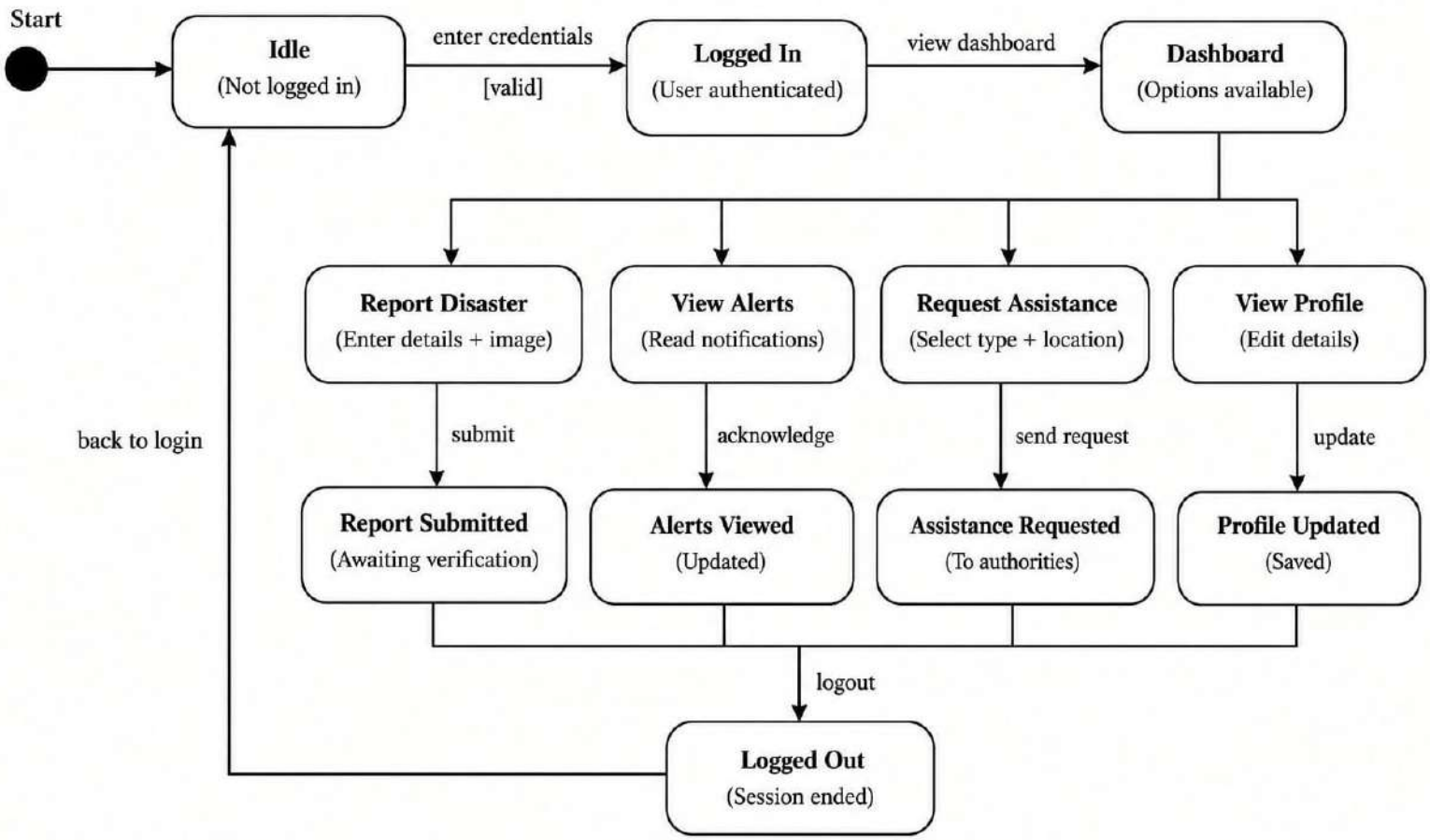

Disaster Management System
Login
Logout
User Use Cases
Register
Update Profile
Report Disaster
Upload Photos / Videos
View Disaster Status
View Alerts & Notifications
View Shelter Information
Request Rescue
Send SOS
View Emergency Contacts
<<include>>
Upload Photos / Videos
<<include>>
Share Location
<<include>>
<<extend>>
Get Current Location
Admin Use Cases
Verify Disaster Reports
Manage Users
Manage Rescue Teams
Manage Resources
Manage Shelters
Send Emergency Alerts
Generate Reports
View Dashboard
User
Admin
<<include>> : Mandatory inclusion of use case
<<extend>> : Optional extension of use case

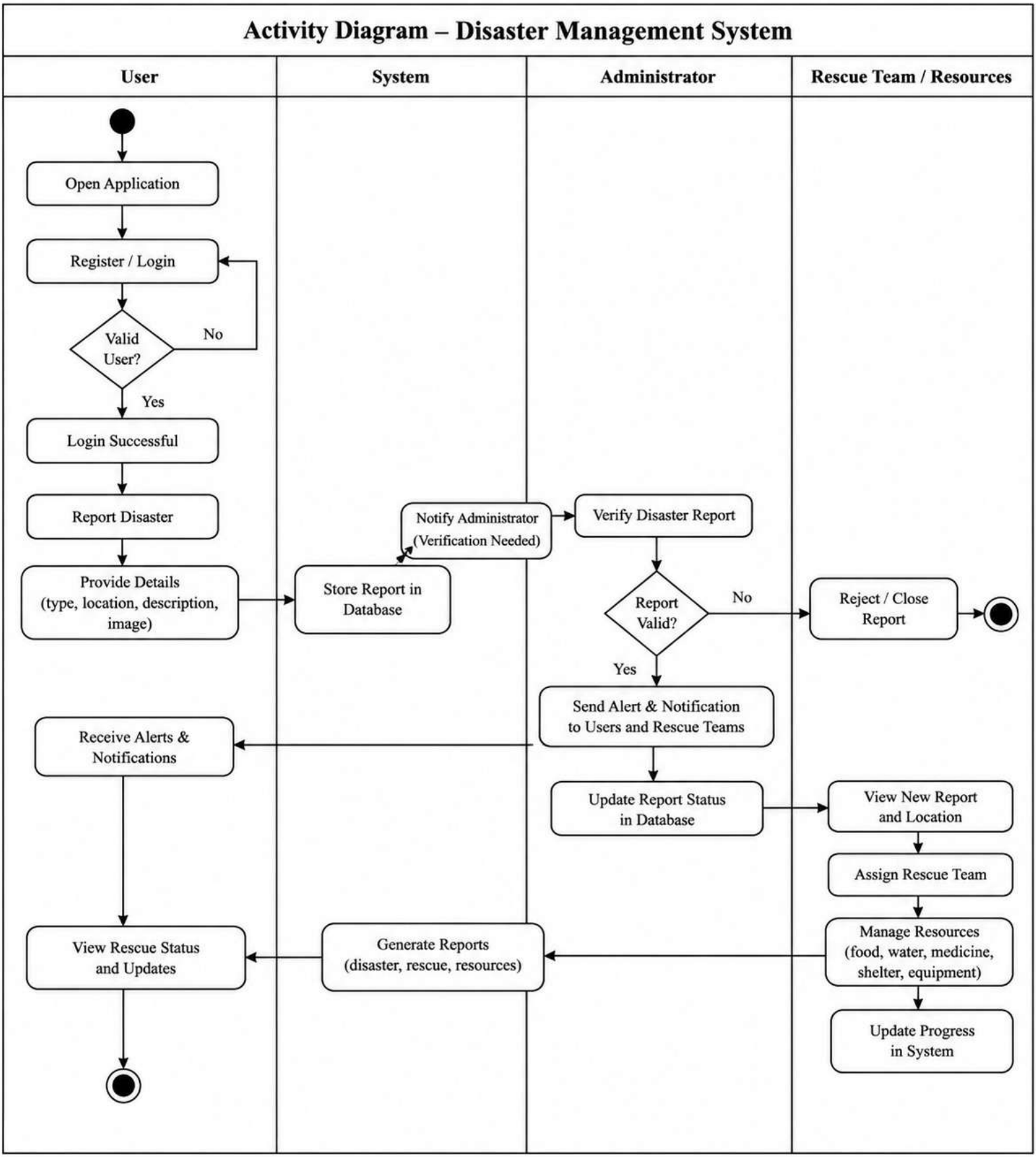

Disaster Management System – Class Diagram
DisasterReport
- reportID : int
- userID : int
- type : string
- location : string
- description : string
- dateTime : datetime
- status : string
- photo : string
+ createReport()
+ updateReport()
+ deleteReport()
+ viewReport()
reports
User
- userID : int
- name : string
- email : string
- password : string
- phone : string
- address : string
- role : string
- registrationDate : datetime
+ register()
+ login()
+ updateProfile()
+ logout()
managed by
Admin
- adminID : int
- name : string
- email : string
- password : string
- phone : string
+ login()
+ manageUsers()
+ verifyReports()
+ manageResources()
+ sendAlerts()
+ generateReports()
generates
Alert
- alertID : int
- reportID : int
- message : string
- alertType : string
- dateTime : datetime
- status : string
+ createAlert()
+ sendAlert()
+ viewAlert()
requests
RescueRequest
- requestID : int
- userID : int
- reportID : int
- location : string
- requestTime : datetime
- status : string
+ createRequest()
+ updateStatus()
+ viewRequest()
assigned to
RescueTeam
- teamID : int
- name : string
- leader : string
- contactNo : string
- specialization : string
- status : string
+ addTeam()
+ updateTeam()
+ viewTeam()
manages
uses
Resource
- resourceID : int
- name : string
- type : string
- quantity : int
- location : string
- status : string
+ addResource()
+ updateResource()
+ viewResource()
manages
gives
Feedback
- feedbackID : int
- userID : int
- message : string
- rating : int
- dateTime : datetime
+ submitFeedback()
+ viewFeedback()
allocates
Shelter
- shelterID : int
- name : string
- location : string
- capacity : int
- availableBeds : int
- contactNo : string
+ addShelter()
+ updateShelter()
+ viewShelter()

Sequence Diagram – Disaster Reporting

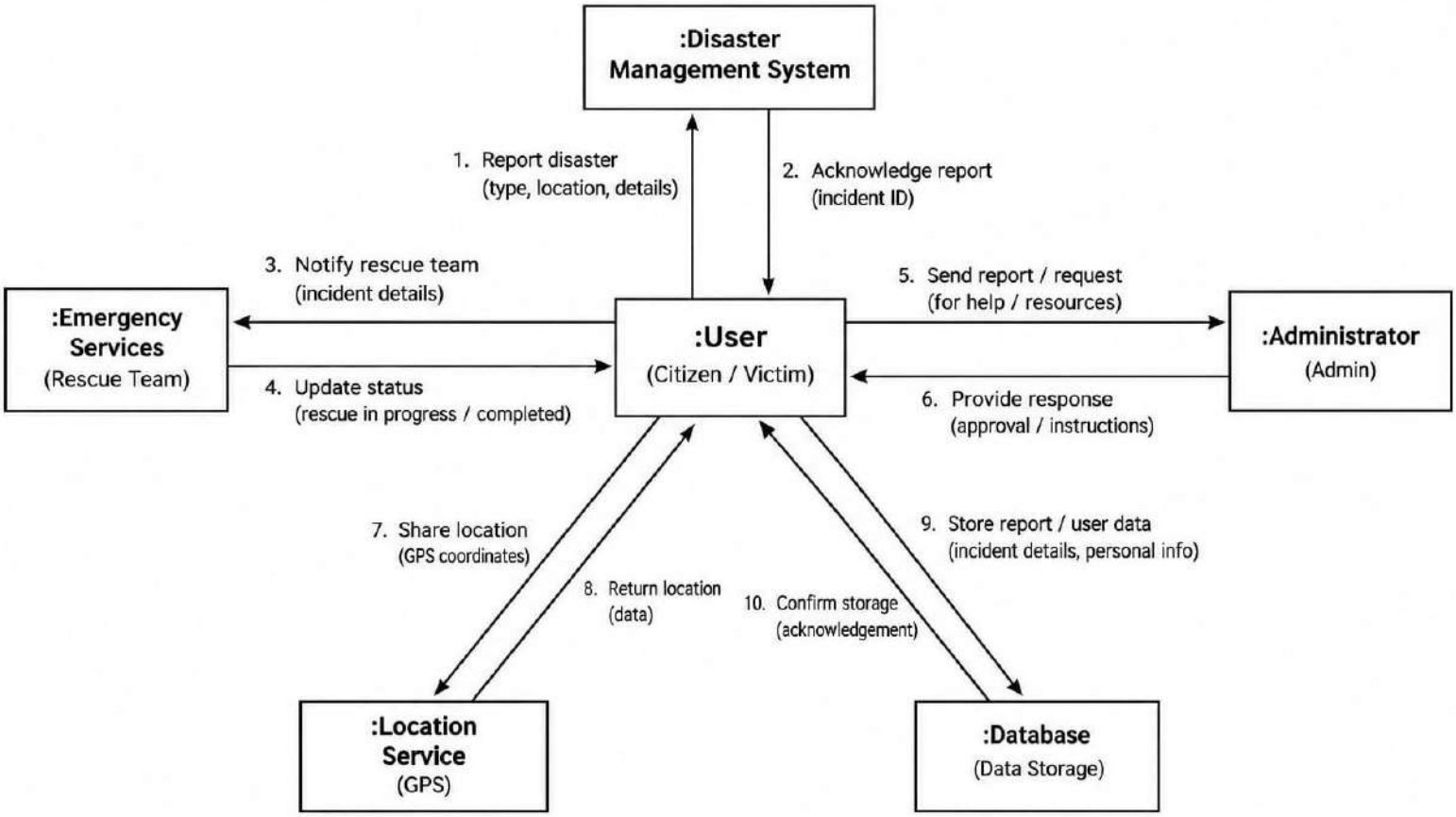


State Diagram – User (Disaster Management System)

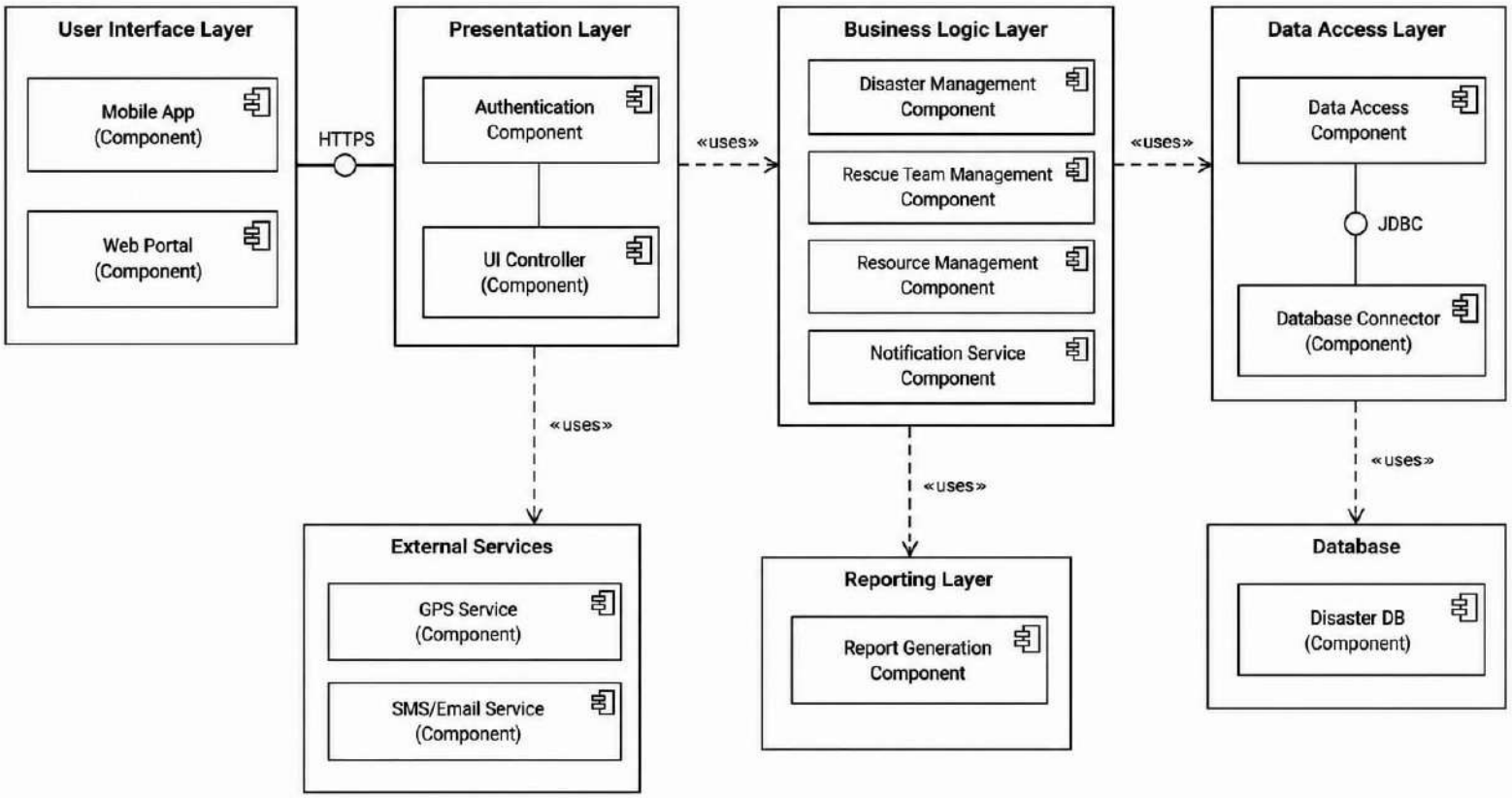


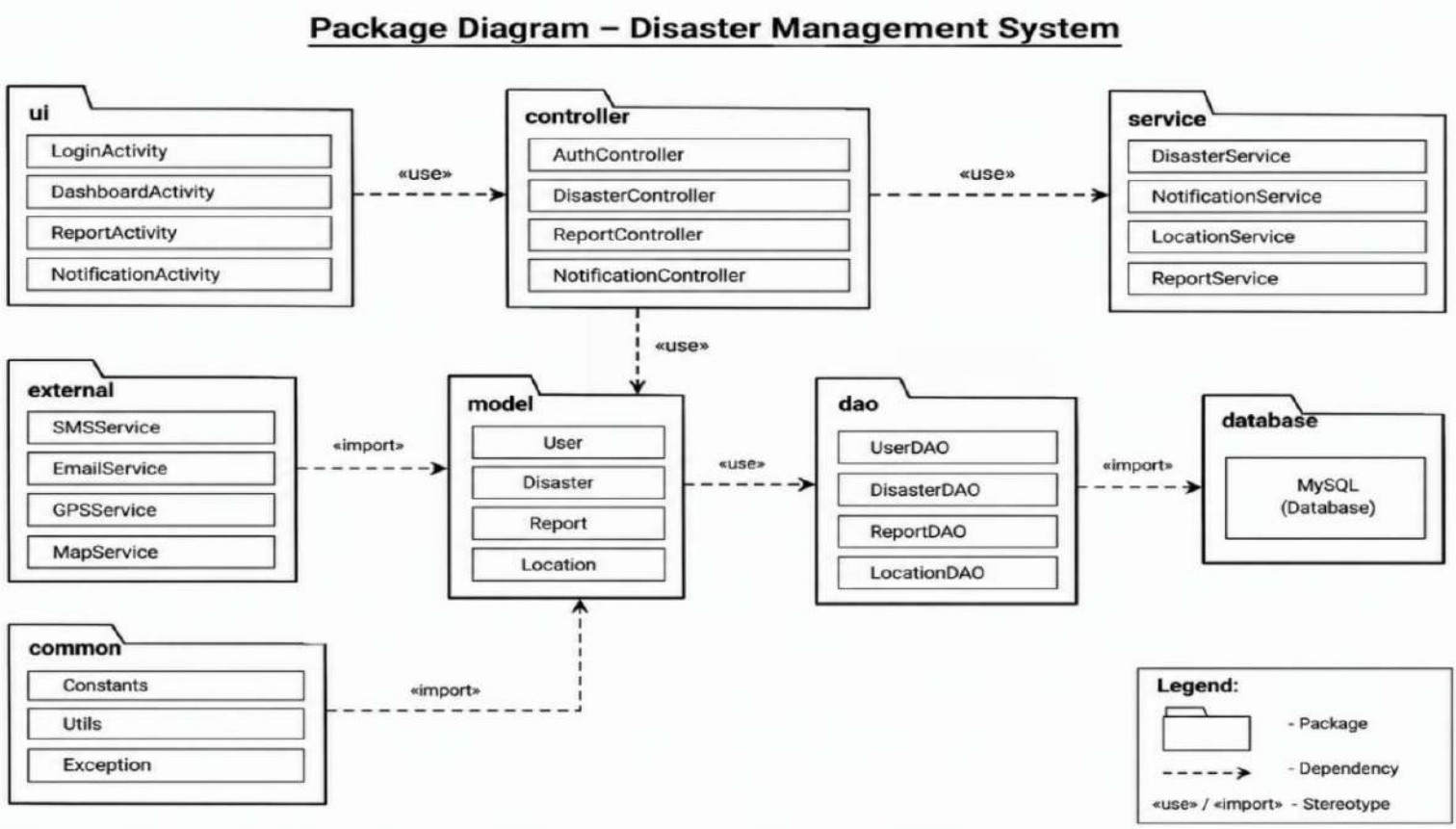

Activity Diagram – Disaster Management System
User
System
Administrator
Rescue Team / Resources
Open Application
Register / Login
Valid User?
No
Yes
Login Successful
Report Disaster
Provide Details (type, location, description, image)
Store Report in Database
Notify Administrator (Verification Needed)
Verify Disaster Report
Report Valid?
No
Reject / Close Report
Yes
Send Alert & Notification to Users and Rescue Teams
Receive Alerts & Notifications
Update Report Status in Database
View New Report and Location
Assign Rescue Team
Manage Resources (food, water, medicine, shelter, equipment)
Update Progress in System
Generate Reports (disaster, rescue, resources)
View Rescue Status and Updates

Collaboration Diagram – Disaster Management System



Component Diagram – Disaster Management System




Package Diagram – Disaster Management System
ui
LoginActivity
DashboardActivity
ReportActivity
NotificationActivity
«use»
controller
AuthController
DisasterController
ReportController
NotificationController
«use»
service
DisasterService
NotificationService
LocationService
ReportService
«use»
external
SMSService
EmailService
GPSService
MapService
«import»
model
User
Disaster
Report
Location
«use»
dao
UserDAO
DisasterDAO
ReportDAO
LocationDAO
«import»
database
MySQL
(Database)
common
Constants
Utils
Exception
«import»
Legend:
- Package
- Dependency
«use» / «import» - Stereotype

Deployment Diagram – Disaster Management System

